

Lincoln University College**Assignment Title: Task Analysis of an Interactive System****Total Marks: 15****Assignment Question**

Select one user task from any interactive system, mobile application, web application, or social media platform that is commonly available in Nepal. Perform a detailed task analysis of the selected activity and suggest user-centered design improvements.

You may choose any one of the following use cases:

1. Withdrawing money from an ATM
2. Transferring money using a mobile banking app
3. Paying an electricity, water, internet, or mobile bill online
4. Recharging a mobile phone using a digital wallet
5. Booking a movie ticket online
6. Ordering food through Foodmandu, Bhoj, Pathao Food, or similar apps
7. Booking a ride using Pathao, inDrive, or similar apps
8. Shopping for products on Daraz Nepal or other e-commerce websites
9. Tracking an online order delivery
10. Booking a domestic flight ticket online
11. Checking in for a flight using an airline website or app
12. Booking a bus ticket online
13. Applying for a passport appointment online
14. Filling out a driving license application form online
15. Checking driving license exam results online
16. Paying vehicle tax or renewal fees online
17. Registering for a course through a college or university portal
18. Submitting assignments through Google Classroom, Veda app, Moodle, or Microsoft Teams

19. Attending an online class using Zoom, Google Meet, Veda app, or Microsoft Teams
20. Sending an email with an attachment
21. Creating or updating a CV on an online job portal
22. Applying for a job through MeroJob, JobsNepal, LinkedIn, or similar platforms
23. Booking a doctor's appointment through a hospital website or app
24. Viewing medical reports through a hospital online system
25. Using a library management system to search or renew books
26. Filing a complaint or service request through a government or organization website
27. Checking exam results through an online result portal
28. Creating a new account on Facebook, Instagram, TikTok, or LinkedIn
29. Posting text, photos, videos, stories, reels, short videos, or live sessions on a social media platform
30. Commenting, liking, sharing, or saving content on social media
31. Sending a message or media file using Messenger, WhatsApp, Viber, or Telegram
32. Reporting inappropriate content or blocking a user on social media
33. Creating and managing a group, page, or event on Facebook or similar platforms
34. Using a web app such as Google Docs, Canva, Trello, Notion, GitHub, or ChatGPT to complete a task

Assignment Requirements and Marking Scheme

S.N.	Criteria	Description	Marks
1	Description of the selected task and target users	Explain the selected task and describe the main users who perform this task.	2
2	Main user goal and sub-goals	Identify the main goal of the user and divide it into smaller sub-goals.	2
3	Step-by-step task analysis	Present the steps involved in completing the task using a table, flowchart, or both.	3

S.N.	Criteria	Description	Marks
4	Screenshots, figures, or interface diagrams	Include relevant screenshots, figures, or interface diagrams showing important stages of the task. Each should be properly labeled and briefly explained.	2
5	Identification of possible user difficulties	Discuss problems users may face, such as unclear instructions, confusing buttons, too many steps, slow loading, errors, security concerns, language barriers, or accessibility issues.	3
6	User-centered design suggestions	Suggest improvements that would make the system easier, faster, safer, more accessible, and more user-friendly.	3
Total			15

Expected Output

The assignment should include the following:

1. A brief explanation of the selected task and target users
 2. Main user goal and sub-goals
 3. Task analysis table
 4. Flowchart of the task process
 5. Labeled screenshots, figures, or interface diagrams
 6. Explanation of possible user difficulties
 7. User-centered design improvement suggestions
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Submission Guidelines

- The assignment should be written in a clear and organized format.
- Use proper headings and subheadings.
- Screenshots, figures, and flowcharts must be clearly labeled.
- The work should be original and based on the selected user task.
- Submit the assignment as instructed by the course lecturer or department.

LINCOLN UNIVERSITY COLLEGE

Faculty of Information Technology

Human Computer Interaction

Assignment 1

Task Analysis of an Interactive System

Selected Task:

Transferring Money Using a Mobile Banking App

Submitted To Er. Mithlesh Mishra Lecturer, Human Computer Interaction Lincoln University College	Submitted By Name: _____ Roll No.: _____ Program / Semester: _____
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Date of Submission: June 20, 2026

1. Description of the Selected Task and Target Users

1.1 The Selected Task

The selected user task is transferring money from one's own bank account to another person's bank account using a smartphone mobile banking application. This is one of the most frequently performed digital transactions in Nepal today. Customers no longer need to visit a branch or an ATM; instead they open the bank's mobile app, enter the recipient's details and the amount, authorise the payment with a PIN or OTP, and receive an instant digital receipt.

In the Nepali context this task is carried out through bank-issued applications such as NIC Asia Mobile Banking, Nabil mBank, Global IME Mobile Banking and Siddhartha Bank Smart, as well as digital wallets like eSewa and Khalti. Inter-bank transfers are settled in the background through national payment rails such as connectIPS and the NPI network. For this analysis a representative app is used and referred to as “SmartBank Mobile” to keep the discussion general while remaining true to how these apps actually behave.

1.2 Target Users

The task is performed by a broad cross-section of bank customers who own a smartphone and have activated mobile banking. The main user groups include:

- **Students and young adults** who pay tuition fees, hostel rent and split bills with friends.
- **Salaried employees** who transfer money to family, repay loans or pay landlords each month.
- **Small business owners and shopkeepers** who settle payments with suppliers and accept money from customers.
- **Parents and remittance receivers** often middle-aged or elderly, sending money to children studying in another city.

These users vary widely in age, income, education and digital literacy. Some are confident smartphone users, while others are first-time or occasional users who feel anxious about making mistakes with their money. Many are more comfortable reading Nepali than English, and a number have limited eyesight or use the app on low-cost phones with unstable internet. A good design must therefore serve experts and beginners equally well.

2. Main User Goal and Sub-Goals

2.1 Main Goal

The main goal of the user is to securely and accurately transfer a specific amount of money from their own account to the correct recipient's account through the mobile app, and to receive confirmation that the transfer was successful.

2.2 Sub-Goals

To achieve this main goal, the user must complete a series of smaller sub-goals. These can be expressed as a Hierarchical Task Analysis (HTA):

1. **Gain secure access** — open the app and authenticate using MPIN, password or fingerprint.
2. **Reach the transfer function** — locate and open the ‘Fund Transfer’ option from the home screen.

3. **Provide transfer details** — choose the source account and enter the beneficiary's bank, account number, name and the amount.
4. **Verify and authorise** — review the confirmation summary and approve it with an OTP or transaction PIN.
5. **Confirm and keep a record** — view the success message and save or share the digital receipt.

3. Step-by-Step Task Analysis

The table below breaks the task into sequential steps, showing the action the user performs and the response the system gives at each stage. The accompanying flowchart (Figure 6) presents the same process visually, including the decision points where the task can branch or repeat.

3.1 Task Analysis Table

Step	User Action	System Response
1	Open the SmartBank Mobile app	Displays the login screen
2	Enter MPIN / password or tap fingerprint	Authenticates the user and validates credentials
3	(If valid) lands on home dashboard	Shows account balance and quick-action menu
4	Tap the 'Fund Transfer' option	Opens the fund transfer form
5	Select the source account	Loads available balance for that account
6	Enter beneficiary bank, account number and name	Validates account number format; may fetch account name
7	Enter the amount and optional remarks	Checks the amount against balance and limits
8	Tap 'Proceed'	Displays a confirmation summary of the transfer
9	Review details and request OTP / enter transaction PIN	Sends an OTP by SMS or prompts for the PIN
10	Enter the OTP / PIN and tap 'Confirm & Pay'	Verifies the code and processes the transaction
11	Wait for processing	Debits the account and completes the transfer
12	View the success screen	Shows a digital receipt with a transaction ID
13	Save or share the receipt	Generates a shareable / downloadable receipt

3.2 Flowchart of the Task Process

Figure 6 shows the complete flow, including the two key decision points: whether login is successful and whether the OTP is valid and the balance is sufficient. If either check fails, the user is looped back to correct the input before continuing.

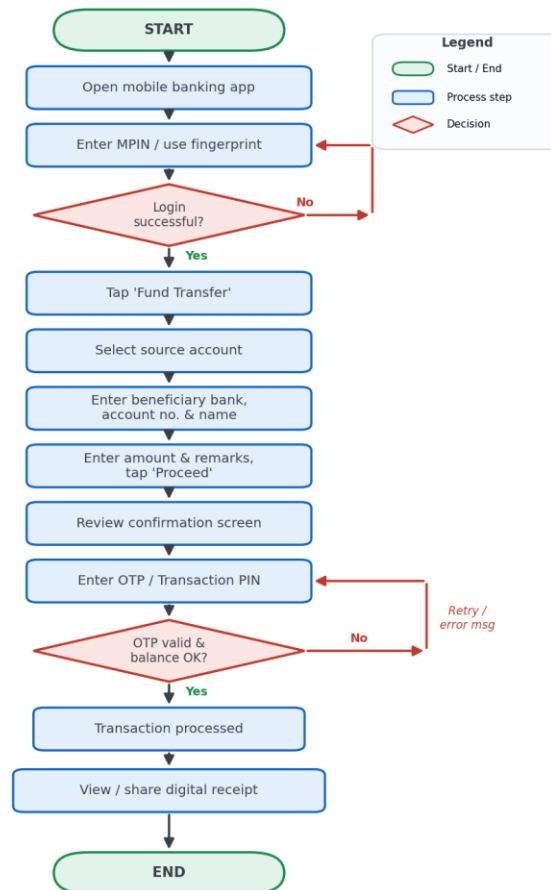


Figure 6: Flowchart of the money-transfer task, showing process steps and decision points.

4. Interface Diagrams of Key Stages

The following labelled interface diagrams illustrate the five most important stages a user passes through. They are simplified wireframes of a typical Nepali mobile banking app and are used here in place of live screenshots so that no real account data is shown.



Figure 1: Login / authentication screen

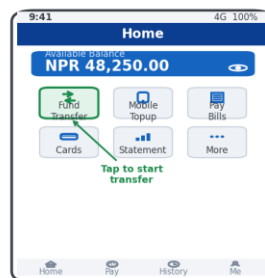


Figure 2: Home dashboard with 'Fund Transfer'

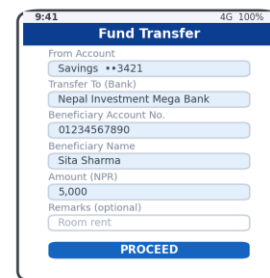


Figure 3: Fund transfer entry form

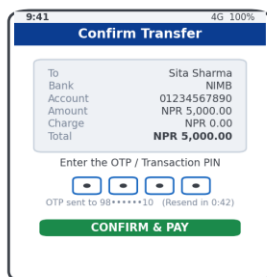


Figure 4: Confirmation summary and OTP entry

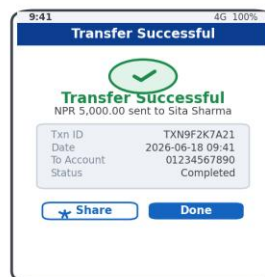


Figure 5: Success screen / digital receipt

Explanation of the Stages

- **Figure 1 – Login:** the user authenticates with an MPIN/password or fingerprint before any account information is shown.
- **Figure 2 – Dashboard:** the balance and a grid of quick actions appear; 'Fund Transfer' is the starting point of the task.
- **Figure 3 – Transfer form:** the user supplies the source account, beneficiary details and amount.
- **Figure 4 – Confirmation & OTP:** a summary lets the user check everything before authorising with a one-time code.
- **Figure 5 – Success:** a receipt with a transaction ID confirms the transfer and can be shared or saved.

5. Possible User Difficulties

Despite the convenience of mobile transfers, users in Nepal face a number of practical difficulties. The table below groups the most common problems with concrete examples.

Type of Difficulty	Description and Example
Confusing layout / buttons	Quick-action icons are not always clearly labelled, so beginners are unsure which option starts a transfer versus a top-up or bill payment.
Too many steps	First-time transfers often require registering the beneficiary, waiting a cooling-off period, then transferring — a long, repetitive flow that frustrates users.
Unclear error messages	Errors such as 'Transaction failed (E-301)' give a code but no plain explanation or guidance on what to do next.
Slow loading / network issues	On weak mobile data the confirmation or OTP screen may freeze, leaving the user unsure whether money was actually deducted.
OTP and timing problems	SMS OTPs arrive late or the entry timer expires, forcing the user to restart and re-enter all details.
Security concerns	Users worry about sending money to a wrong or fraudulent account, especially when the beneficiary name is not shown for verification.
Language barrier	Interfaces are mostly in English; older users and those more comfortable in Nepali struggle with terms like 'beneficiary' or 'remarks'.
Accessibility issues	Small fonts, low colour contrast and no screen-reader support make the app hard to use for elderly users and those with low vision.

6. User-Centered Design Suggestions

The improvements below are mapped directly to the difficulties above. They aim to make the transfer faster, clearer, safer and more inclusive, and are grounded in well-known usability principles such as Nielsen's heuristics.

Problem Addressed	Suggested Improvement	Main Benefit
Confusing buttons	Use clear text labels with familiar icons and highlight the primary action; group transfer-related options together.	Easier to learn
Too many steps	Offer a saved-beneficiaries / recent-recipients list so repeat transfers take two taps instead of re-entering everything.	Faster
Unclear errors	Replace error codes with plain-language messages that say what went wrong and how to fix it.	Less frustration
Network / slow loading	Show clear progress states and a safe retry; queue the request and confirm the final status to avoid double payments.	More reliable
OTP problems	Allow biometric or in-app approval instead of SMS OTP, with a generous timer and an easy 'resend'.	Smoother authorisation
Security worries	Display the verified beneficiary name before confirmation and mask account numbers; add a clear 'review' summary.	Safer, builds trust
Language barrier	Provide a one-tap Nepali / English language toggle using simple everyday words.	More inclusive
Accessibility	Support larger fonts, high-contrast mode and screen readers; ensure tap targets are big enough.	Accessible to all

Summary of the User-Centered Approach

Taken together, these changes shorten the task, reduce the chance of costly mistakes, reassure users about security and open the app to people of different ages, languages and abilities. The guiding idea of user-centered design is to fit the system to the user rather than forcing the user to adapt to the system — keeping each step visible, forgiving of errors and easy to understand.

7. Conclusion

Transferring money through a mobile banking app is a routine but high-stakes task for millions of users in Nepal. The task analysis shows that the activity is logically simple — authenticate, enter details, authorise, confirm — yet several friction points around navigation, error handling, OTPs, language and accessibility can make it stressful, especially for less experienced users. By applying user-centered design improvements such as saved beneficiaries, plain-language feedback, name verification, biometric approval and bilingual, accessible interfaces, banks can make the transfer experience faster, safer and friendlier for everyone.

8. REFERENCES

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